Design a web dashboard using Flask. Control relay module via web.

Installation Manual

Hardware Requirements

1. Raspberry Pi Model A/B/B+
2. Jumper wires (Female to Male)
3. Relay Module

- **Software Requirements**
  1. Raspbian Stretch OS

1. **Connect your Relay Module with Raspberry Pi's GPIO Pins.**

Relay	Relay Input	RPI Physical Pin	Raspberry Function
VCC	POWER	2	VCC
GND	GND	6	GND
CH 1	IN 1	12	GPIO 18

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- **Code:-**
- **1. Python code:**

```
from flask import Flask, render_template, redirect, url_for
import RPi.GPIO as GPIO

app = Flask(__name__)

RELAY = 12

GPIO.setmode(GPIO.BCM)
GPIO.setup(RELAY, GPIO.OUT)

relay_state = False
GPIO.output(RELAY, GPIO.HIGH)

@app.route("/")
def index():
    return render_template("index.html", state=relay_state)

@app.route("/on")
def turn_on():
    global relay_state
    GPIO.output(RELAY, GPIO.LOW)
    relay_state = True
    return redirect(url_for("index"))

@app.route("/off")
def turn_off():
    global relay_state
    GPIO.output(RELAY, GPIO.HIGH)
    relay_state = False
    return redirect(url_for("index"))

if __name__ == "__main__":
    try:
        app.run(host="0.0.0.0", port=5000)
    finally:
        GPIO.cleanup()
```

2. **HTML code**

Write a code in idle and save as index.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Relay Control Dashboard</title>
    <meta name="viewport" content="width=device-width, initial-scale=1">

    <style>
        * {
            box-sizing: border-box;
        }

        body {
            margin: 0;
            font-family: Arial, sans-serif;
            background: #f1f5f9;
            display: flex;
            justify-content: center;
            align-items: center;
            min-height: 100vh;
        }

        .dashboard {
            width: 90%;
            max-width: 450px;
            background: white;
            padding: 35px;
            border-radius: 20px;
            text-align: center;
            box-shadow: 0 8px 25px rgba(0,0,0,0.15);
        }

        h1 {
            margin-bottom: 5px;
        }

        .subtitle {
            color: #666;
            margin-bottom: 30px;
        }
```

```
        .relay-box {
            padding: 25px;
            border-radius: 15px;
            background: #f8fafc;
        }

        .status {
            font-size: 24px;
            font-weight: bold;
            margin: 20px;
        }

        .on-status {
            color: green;
        }

        .off-status {
            color: red;
        }

        .button {
            display: block;
            text-decoration: none;
            color: white;
            padding: 15px;
            margin: 12px 0;
            border-radius: 10px;
            font-size: 20px;
            font-weight: bold;
        }

        .on-button {
            background: green;
        }

        .off-button {
            background: red;
        }
    </style>
```

```
</head>

<body>

<div class="dashboard">

    <h1>Raspberry Pi</h1>
    <div class="subtitle">Relay Control Dashboard</div>

    <div class="relay-box">

        <h2>Relay 1</h2>

        {% if state %}
            <div class="status on-status">● ON</div>
        {% else %}
            <div class="status off-status">● OFF</div>
        {% endif %}

        <a href="/on" class="button on-button">TURN ON</a>

        <a href="/off" class="button off-button">TURN OFF</a>

    </div>

</div>

</body>
</html>
```

To Run the Programme:

Please note:- Before updating terminal you have to save your python programme in the folder of "relay" and "html code" inside the folder of template.

1. To get host address

1. **Open the terminal**

   Write a command (hostname -I)

2.Open the terminal

1. ***mkdir relay (Create a folder)***
2. ***cd relay***
3. ***mkdir template (Create another folder inside the relay folder)***

3.Then open another terminal

1. ***sudo apt update***
2. ***sudo apt install python3-flask***
3. ***python3 –c "import flask ; print ('Flask OK')"***
4. ***python3 app.py***
5. ***Then copy url and paste in chrome***